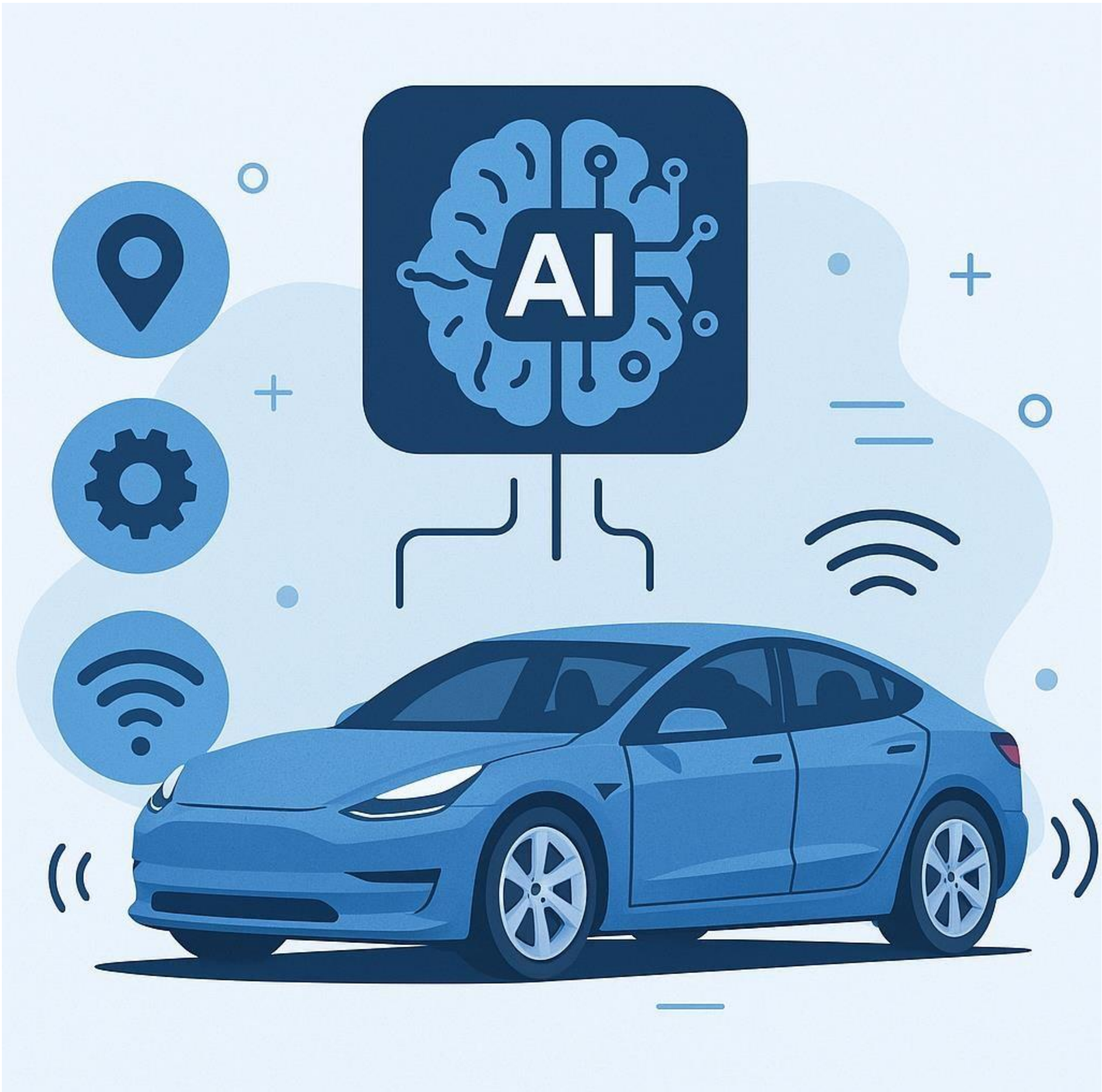

Eyes on ~~the Road~~…YOU?

Privacy in the age of Advanced Technology in Cars

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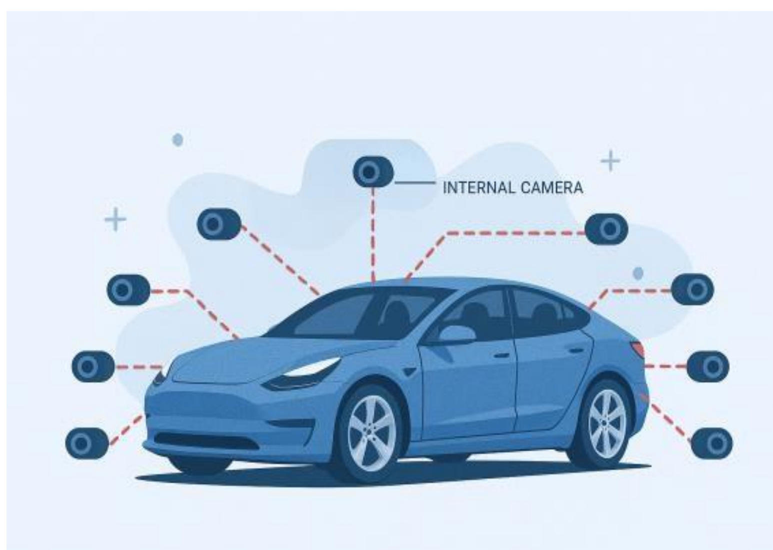
It’s Officially the Future

Fully self driving cars used to be a figment of the imagination, something you might see in Science Fiction movies and think, *Wow! Maybe in 2030!* But with the rapid development of Artificial Intelligence in the past decade, self driving cars have very much become a reality. While AI has the potential to revolutionize the automotive industry, there are plenty of risks hidden behind the excitement of innovation along for the ride.

Contrary to popular belief, self driving cars are far from a new idea. Fully autonomous vehicles date back to the 16th century, “when Leonardo da Vinci designed a small, three-wheeled, self-propelled cart regarded as not only the first self-driving vehicle, but the first robot of any kind”[1]. Centuries later, technology has caught up to the ideas of the greats, and fully autonomous vehicles are on the road in major US cities like Los Angeles, Phoenix, and San Francisco, offering taxi services to civilians.

These services have developed rapidly and the public has become blindsided by the glamor of having a car with an iPad that you can play videogames on. AI integration in vehicles means more than just features like fart noises, funny horn sounds, and light shows. As Tesla and other autonomous car companies continue to technologically advance automotive vehicle industry, its crucial to consider the potential dangers regarding data privacy, and how individuals can protect themselves from these risks.

Smile for the Camera!



Although fully self driving cars are not available for American consumers, yet, there are some on the market with the ability to operate with little to no human assistance. Tesla has implemented aspects of full self driving and autopilot, a proprietary driver assistance system, using a unique exclusively camera vision and AI model based system. This system uses image recognition in an attempt to drive like a human with nine eyes. All Tesla models come with eight external cameras, these are the eyes on the road playing a crucial role in autopilot features like, identifying obstacles, collision avoidance, and security when in park. Additionally there is one internal facing camera, Tesla’s website states, “The cabin camera can determine driver inattentiveness and provide you with audible alerts, to remind you to keep your eyes on the road when Autopilot is engaged. [2]" A live stream from these cameras can be seen when the car is in park by key holders in the Tesla app. These cameras work hand in hand with the AI technology to implement split second image recognition to make decisions and control the vehicle.

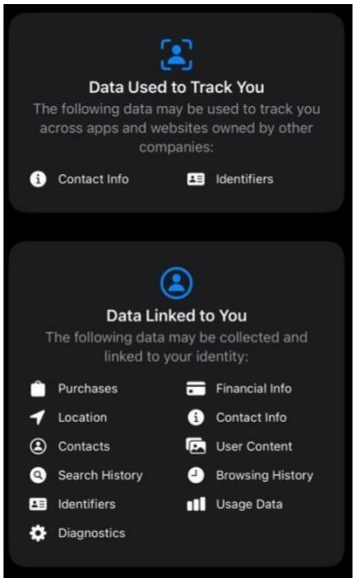
Tesla's privacy policy clearly states that Tesla does not rent or sell the data collected to third parties, but this does not mean that no one will ever see that video of you slipping and falling on the ice behind a neighbors Model 3. These cameras provide a feature called sentry mode, essentially an advanced dashcam that records a 360 degree view of the car. The video recordings are saved to a hard drive in the vehicle leaving it up to drivers to decide whether or not they want to keep the video. Videos will automatically be deleted to free up space on the hard drive, but drivers are able to select videos to be saved and downloaded to any key-device. This leaves car owners in possession of countless videos of bystanders obtained without their consent creating virtually limitless potential privacy issues. Or is it just drivers?

While vehicle owners may decide to save or delete sentry mode footage, in certain situations, Tesla itself stores data obtained from the vehicles. Tesla's 360 degree view becomes a potential asset when something goes wrong. For this reason, camera recordings, location data, and speed data are, "Shared with Tesla and associated with your account, In the occurrence of a safety critical event only.[2]" A safety critical event can mean anything from the car has recordings of a crime being committed, an accident involving the car itself, and other incidents of that nature. Although these situations are rare and it has proven to be beneficial, concerns arise as your data is being stored somewhere leaving it vulnerable to hacking and misuse. As a whole it is critical to keep data privacy in mind as we equip vehicles with more sensors, cameras, and access to personal data.

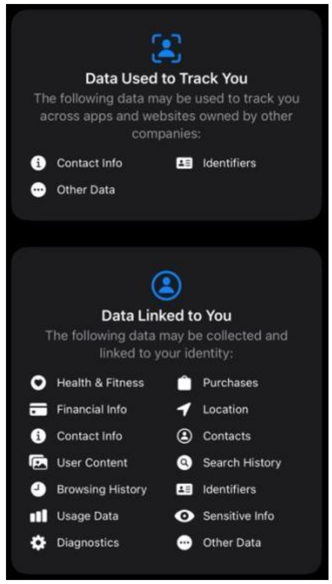
Tesla's NOT the Only One Watching

It is clear that car companies having access to personal data can leave individuals feeling vulnerable. As shown in a survey provided by the PEW Research Center, 76% of US adults believe that "The computer systems in driverless passenger vehicles would be easily hacked in ways that put safety at risk.[3]" While unfortunately there is always a risk of hacking and misuse when it comes to technology, it is important to identify the current state of individuals data privacy. Phones have a concerning amount of data on individuals. Social media apps have a bad reputation when it comes to protecting personal data. Below are the lists of how users personal data will be used for three of the most popular social media platforms, Facebook, Tik Tok, and Snapchat, as displayed in the iPhone app store.

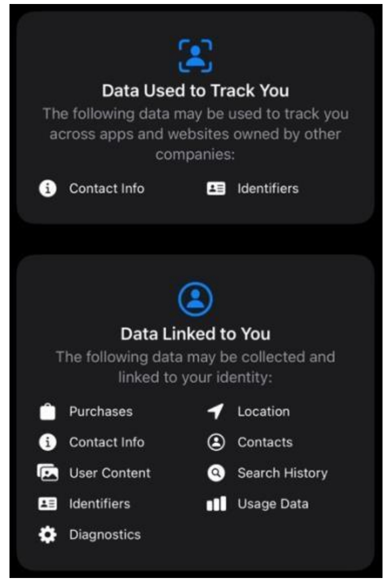
Tik Tok



Facebook



Snapchat



Based on the information shared by social media company's, it is clear to see that users of these platforms data is not protected. It is being collected, used, and sold daily to third party sources. Facebook specifically seems to collect a copious amount of data and has been in countless lawsuits over privacy concerns. They collect data on all facet of users accounts from the type of content interacted with, messages sent and received, and diving into device data tracking its characteristics, information on network connections, and what each individual is doing on the device.

This is an extreme level of invasion of privacy that most individuals experience. All apps and websites preform some level of data collection on users making it borderline impossible to avoid in todays day and age. If this level of tracking and data collection is happening with the devices glued to American's hands, why should it matter if its implemented in cars as well?

The answer to that question lies in the fact that the data collected by vehicles is unlike the data collected by a smartphone. It is not only digital, it is physical, tracking where we go, how fast we drive, who is nearby, and even what happens around us. Phones can be turned off or left behind, but cars can be unavoidable and essential for individuals. The integration of more advanced computer systems in cars extends surveillance from our online presence to our physical movements, expanding the risk to personal privacy in every day life.

Protect Yourself!

For all the Tesla drivers out there it is important to protect your data! Here are a few steps to take, directly from the Tesla Owners Manual to ensure that your data is not being shared with Tesla without your knowledge.

- To adjust all data sharing preferences, on the screen in the car, navigate to…

  Controls > Software > Data Sharing

  From here you can view all data sharing options, it is strongly encouraged to disable them

- To view features currently enabled using the cabin camera, on the screen in the car, navigate to…

  Controls > Software > Cabin Camera

- The car collects browser data similar to a smart phone or computer, to clear browser data, on the screen in the car, navigate to…

  Controls > Service > Clear Browser Data

Sources:

[1] "A Brief History of Autonomous Vehicles – from Renaissance to Reality: Mobileye Blog." *Mobileye*, www.mobileye.com/blog/history-autonomous-vehicles-renaissance-toreality/. Accessed 03 Apr. 2025.

[2] "Owner's Manual." *Tesla*, www.tesla.com/ownersmanual. Accessed 03 Apr. 2025.

[3] Lee Rainie, Cary Funk. "4. Americans Cautious about the Deployment of Driverless Cars." *Pew Research Center*, 17 Mar. 2022, www.pewresearch.org/internet/2022/03/17/americans-cautious-about-the-deployment-ofdriverless-cars/. Accessed 03 Apr. 2025.

[4] "*Privacy Not Included: A Buyer's Guide for Connected Products." *Mozilla Foundation*, foundation.mozilla.org/en/privacynotincluded/articles/its-official-cars-arethe-worst-product-category-we-have-ever-reviewed-for-privacy/. Accessed 03 Apr. 2025.

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